

EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.
DATE:26/03/2025	

AIM:

To develop an Online Quiz Application using Java Servlets and demonstrate session tracking using HttpSession, where a user logs in, attempts a quiz, and their session is maintained throughout the interaction.

ALGORITHM:

1. Input:

- The user enters username and password in the login form.
- After login, the user attempts quiz questions.
- The servlet validates login and tracks user session during the quiz.

2. Steps for Implementing Session Tracking:

Step 1: Create Login Form (HTML)

- Create an HTML page with:
 - Username field
 - Password field
 - Login button

Step 2: Create Login Servlet

- Retrieve username and password using request.getParameter().
- Validate credentials (hardcoded).
- If valid:
 - Create session using HttpSession.
 - Store username using session.setAttribute().
- If invalid:
 - Display error message.

Step 3: Redirect to Quiz Page

- After successful login:
 - Redirect to quiz servlet/page.
- Check session before allowing access.

Step 4: Create Quiz Page

- Display multiple-choice questions.
- Use POST method to submit answers.

Step 5: Evaluate Quiz (Servlet)

- Retrieve answers using request.getParameter().
- Calculate score.
- Store score in session (optional).

Step 6: Maintain Session

- Use session.getAttribute("user") to identify logged-in user.
- Prevent unauthorized access.

Step 7: Logout and Session Handling

- Provide logout option.
- Use session.invalidate() to destroy session.
- Redirect to login page.

Program:

Index.html (Login Page)

```
<!DOCTYPE html>

<html>

<head>

    <title>Quiz Login</title>

</head>

<body style="font-family:Segoe UI;text-align:center;margin-top:100px;">

<h2>Online Quiz Login</h2>

<form action="QuizApp" method="post">

    <input type="text" name="u" placeholder="Username" required><br>

    <input type="password" name="p" placeholder="Password" required><br>

    <button type="submit">Login</button>

</form>

</body>

</html>
```

Quiz.html (Quiz Page)

```
<!DOCTYPE html>

<html>

<head>

    <title>Quiz</title>
```

```
</head>
```

```
<body style="font-family:Segoe UI;">
```

```
<h2>Online Quiz</h2>
```

```
<form action="QuizApp" method="post">
```

```
<p>1. Java is a?</p>
```

```
<input type="radio" name="q1" value="a"> Programming Language<br>
```

```
<input type="radio" name="q1" value="b"> Device<br>
```

```
<p>2. HTML stands for?</p>
```

```
<input type="radio" name="q2" value="a"> Hyper Text Markup Language<br>
```

```
<input type="radio" name="q2" value="b"> High Machine Language<br>
```

```
<p>3. Which method is secure?</p>
```

```
<input type="radio" name="q3" value="a"> GET<br>
```

```
<input type="radio" name="q3" value="b"> POST<br>
```

```
<br>
```

```
<button type="submit">Submit Quiz</button>
```

```
</form>
```

```
</body>
```

```
</html>
```

```
QuizServlet.java
```

```
import java.io.*;
```

```
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
```

```
@WebServlet("/QuizApp")
```

```
public class QuizServlet extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest req, HttpServletResponse res)
        throws IOException {
```

```
        String u = req.getParameter("u");
```

```
        String p = req.getParameter("p");
```

```
        // LOGIN VALIDATION
```

```
        if (u != null && p != null) {
```

```
            if ("admin".equals(u) && "123".equals(p)) {
```

```
                HttpSession session = req.getSession();
```

```
                session.setAttribute("user", u);
```

```
                res.sendRedirect("Quiz.html");
```

```
                return;
```

```
            } else if (req.getSession(false) == null) {
```

```
                res.getWriter().println("Invalid Login");
```

```
                return;
```

```
            }
```

```
        }
```

```
        // QUIZ SUBMISSION
```

```
        HttpSession session = req.getSession(false);
```

```
        if (session != null && session.getAttribute("user") != null) {
```

```
            String q1 = req.getParameter("q1");
```

```

String q2 = req.getParameter("q2");
String q3 = req.getParameter("q3");

int score = 0;

if ("a".equals(q1)) score++;
if ("a".equals(q2)) score++;
if ("b".equals(q3)) score++;

res.setContentType("text/html");
PrintWriter out = res.getWriter();

out.println("<html><body style='font-family:Segoe UI;text-align:center;'>");
out.println("<h2>Quiz Result</h2>");
out.println("<p>Welcome " + session.getAttribute("user") + "</p>");
out.println("<p>Your Score: <b>" + score + "/" + 3 + "</b></p>");
out.println("<a href='QuizApp?logout=true'>Logout</a>");
out.println("</body></html>");

} else {
    res.sendRedirect("index.html");
}
}

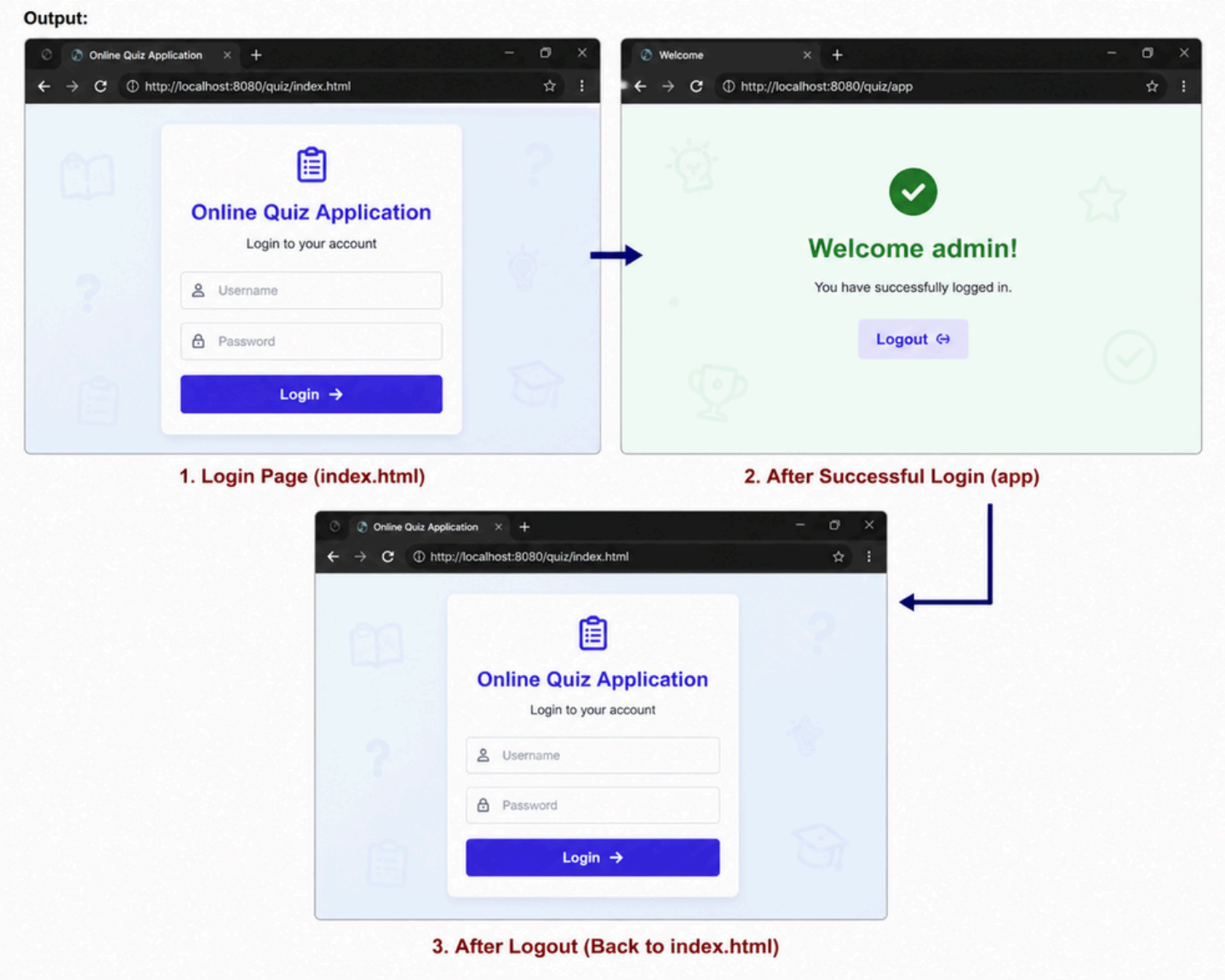
protected void doGet(HttpServletRequest req, HttpServletResponse res)
    throws IOException {

    if (req.getParameter("logout") != null) {
        HttpSession s = req.getSession(false);
        if (s != null) s.invalidate();
        res.sendRedirect("index.html");
    }
}

```

```
}  
}
```

OUTPUT :



RESULT :

An Online Quiz Application is successfully developed using Java Servlets, demonstrating session tracking with `HttpSession`, where user login, quiz access, and result display are managed securely using sessions.

